DATA ANALYSIS INTERNSHIP

DAY 10 REPORT

Matplotlib and Visualization Functions

Introduction

On Day 10, different **data visualization techniques** were explored using **Matplotlib in Google Colab**. Various chart types and plotting functions were practiced to understand how numerical data can be represented visually. Matplotlib helps in presenting complex data in an easy-to-understand graphical format for better interpretation and analysis.

Import Statement

import matplotlib.pyplot as plt

Explanation

- **matplotlib** → Python library used for visualization
- **pyplot** → Module responsible for plotting graphs and charts
- **plt** → Short alias used for easier coding

Common Matplotlib Functions

1. Line Plot

Function: `plt.plot()`

Used to generate line graphs for showing trends or continuous data.

Example:

plt.plot(x, y)

---

2. Bar Chart

Function: `plt.bar()`

Used to compare values through vertical bars.

Example:

plt.bar(x, y)

---

3. Scatter Plot

Function: `plt.scatter()`

Displays the relationship between two variables using points.

Example:

plt.scatter(x, y)

---

4. Histogram

Function: `plt.hist()`

Used to visualize the distribution and frequency of data.

Example:

plt.hist(data)

---

5. Pie Chart

Function: `plt.pie()`

Represents values in percentage form using circular sections.

Example:

plt.pie(values)

---

6. Box Plot

Function: `plt.boxplot()`

Used to display data spread and identify variations.

Example:

plt.boxplot(data)

---

7. Area Plot

Function: `plt.fill_between()`

Used for filling the area between values in a graph.

Example:

plt.fill_between(x, y)

Graph Customization Functions

8. X-axis Label

Function: `plt.xlabel()`

Used to add labels for the x-axis.

---

9. Y-axis Label

Function: `plt.ylabel()`

Used to assign labels for the y-axis.

---

10. Graph Title

Function: `plt.title()`

Adds a suitable title to the graph.

---

11. Legend

Function: `plt.legend()`

Displays labels to identify graph data clearly.

---

12. Grid Lines

Function: `plt.grid()`

Adds grid lines to improve graph readability.

---

13. Display Graph

Function: `plt.show()`

Used to display the plotted graph.

---

14. Figure Size

Function: `plt.figure()`

Adjusts the graph size according to requirements.

Example:

plt.figure(figsize=(5,5))

---

15. Multiple Plots

Function: `plt.subplot()`

Used to place multiple graphs inside a single figure.

---

16. X-axis Range

Function: `plt.xlim()`

Sets the limit for x-axis values.

---

17. Y-axis Range

Function: `plt.ylim()`

Used to define the y-axis range.

---

18. Style Selection

Function: `plt.style.use()`

Applies predefined styles to improve graph appearance.

Applications of Matplotlib

- Data visualization
- Comparing values through charts
- Statistical representation
- Trend and pattern analysis

- Creating reports with visual insights
- Machine learning and analytics projects
- Graphical presentation of datasets

Conclusion

On Day 10, various **Matplotlib plotting and customization functions** were practiced in Google Colab. Different chart types including **line graphs, bar charts, scatter plots, histograms, pie charts, box plots, and area plots** were created to understand data visualization better. In addition, graph customization features such as **labels, titles, legends, grid lines, figure size, and styles** were explored. This session improved understanding of **visual data representation and graph creation in Python**.